# gpeyt1-40 · Wall base pressure

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

## Problem

For a cantilever retaining wall on medium dense sand, the resultant of all vertical forces ΣV = 18,200 lb/ft (Pv included) intersects the 10 ft base at d = 4.1 ft from the toe, computed from moments about the toe. The allowable bearing pressure for the sand is 4,000 psf. Overturning FS = 2.3, sliding FS = 1.6.

**Most nearly, the maximum toe pressure qmax is:**

A. 2,220 psf
B. 2,803 psf
C. 837 psf
D. 1,984 psf

## Given equations / data

> `e = B/2 − d; qmax = (ΣV/B)(1 + 6e/B); qeq (Meyerhof) = ΣV/B′, B′ = B − 2e; for soils e ≤ B/6`

## Worked solution

1. `e = 10/2 − 4.1 = 0.90 ft ≤ B/6 = 1.67 ✓`
2. `qmax = (18,200/10)(1 + 6×0.90/10) = 2,803 psf`
3. `qmin = 837 psf; qeq = 18,200/8.2 = 2,220 psf (for comparison)`

## Final answer

**2,803 psf**

## Why the other values are wrong

- **2,220 psf** — reported Meyerhof uniform qeq
- **837 psf** — reported qmin (heel)
- **1,984 psf** — used (1 + e/B), dropped the 6

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Applied Stress at Base (qmax, qmin, qeq) and eccentricity criteria (cantilever wall figure), p. 209

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=P-W_f8IyVos (no creator values copied).

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